CDL app: deeper moats that build speed can compound

Oct 28, 2026 (PT). Read-only research. [E] = estimate and [J] = judgment; everything else has a cited source. Raw data is in *cdl-2026-10-28/moat/*. This builds on the differentiation, pricing, build-spec and UX reports from today.

0. Bottom line

- **The table-stakes claims are crowded.** 8 of 11 iOS rivals sampled say "all 50 states" in their descriptions. Every major rival lists all six endorsements. Hanh Le already has Spanish and an "Unlimited AI Tutor". Solvely and CDL Pass (Google Play) already have audio. DMV Genie already has a pass guarantee. Claiming any of these in itself is not a moat.
- **What nobody shows is exactness and upkeep.** No sampled rival description mentions:
- Texas's separate Commercial Rules test, which is taken first.
- The three-choice answer format used in New York and Iowa.
- Which states have moved to the AAMVA modernized manual.
- Florida's and Texas's 2026 English-only knowledge tests.
- A public record of fixes.

Meanwhile, the rules keep moving: - 49 CFR Part 383 (CDL standards) had **8 amendment dates in 2026 so far**, and Part 172 (HazMat tables and placards) had **7** (eCFR versioner API). - PHMSA published HM-215R in Feb 2026 and HM-215S on Oct 27, 2026.

A team that ships in days can keep 51 exact state formats current, with a public log. Two-person incumbents demonstrably don't: wrong-answer-key and "not on the test" complaints make up 33% of coded complaints (UX report). - **Speed compounds most in the record, not the features.** The fix log, the miss-rate data, SEO page age and the review count all get better each month and can't be backfilled by a copier. Features themselves (audio, AI, Spanish) can be copied in weeks. - **The ESL moat is weaker than assumed.** - Florida (Feb 6, 2026) and Texas (Jun 1, 2026) now test CDL knowledge in English only. - FMCSA's non-domiciled CDL rule (effective Mar 16, 2026) makes about 194k current holders ineligible over 5 years. - English-proficiency out-of-service enforcement is active.

So reframe it as an "**English-test bridge**": learn in Spanish, pass in English. Don't build it as a Spanish-language test app. - **Odds impact is honest but small.** Moats justify a CDL-specific hit rate of **~0.18 [J] (range 0.14–0.22)** against the queue default of 0.12. In the plan model (Scenario B, C replaces CAL), that moves base odds from **m12 67.1% to 67.5%** and **m18 81.3% to 81.5%**. One app is one draw of about 30. For the CDL app alone, median net profit at maturity goes from **$74 to $131/mo**, and P(≥$1k/mo net) from **11.8% to 17.8%**. Don't lift the hit rate for the rest of the queue on this evidence.

1. Ranked moat stack

Durability runs 1–5, where 5 means hard to erode. Copyability is how fast a funded rival could match it. CC-days = Claude-Code-days of build effort [E].

#	Moat	When	CC-days [E]	Durability	Copyability	Key evidence
1	**Exact-format state engine**: 51 jurisdictions × test list, question count, pass mark, answer choices (3 vs 4), order (TX Commercial Rules first), language rules, manual edition (legacy vs modernized), with a source link and check date per field	Wk 1–2 post-MVP	6–8	4 (upkeep is the moat)	Medium: the facts copy fast, but keeping them current doesn't	TX: Commercial Rules 20 Q / 80%, taken first (TX DPS, 5-Star packet); NY and Iowa use three choices; FMCSA: states write their own tests (≥30 items, 80%); 0 of 11 rival descriptions name these
2	**Change-watch + public fix log**: daily eCFR diff (Parts 383/391/392/393/397/172), Federal Register PHMSA/FMCSA feed, DMV	Wk 1	3–4	5 (a dated history can't be backfilled)	Low	2026 amendment dates: 383 = 8, 172 = 7, 393 = 4, 391 = 3; HM-215R/S; FL and TX English-only; non-domiciled rule;

#	Moat	When	CC-days [E]	Durability	Copyability	Key evidence
	manual page hashes; auto-flags affected questions; dated public log at /fixes					accuracy complaints 33% (n=343)
3	**Programmatic SEO**: state × test pages, each carrying the state's real format table, order, fees and manual date (no spun text)	Wk 2–4, then weekly	4–6	4 (page age and links can't be rushed)	Medium	Semrush: tanker 1,000/mo KD 15; passenger 720 KD 3; CA 720 KD 24; FL 720 KD 12; GA 480 KD 16; TX Commercial Rules 390 KD 28; NY 170 KD 10; doubles/ triples 140 KD 1; school bus 170 KD 8; plus earlier: hazmat 5,400 KD 23, air brakes 880 KD 9, TX 1,000 KD 13
4	**Miss-rate data**: opt-in, aggregate, on-device to anonymous per-question stats; "hardest 25 air-brake questions in Texas" pages; difficulty-calibrated readiness	Month 2 (needs users)	3	4 (grows with users)	Low	Only incumbents with scale have this; none publish it
5	**Test-day topic check-in**: after the test, "which topics came up?", topic chips only, never question text; feeds weighting and a public "what's on the TX test" page	Month 1–2	2	4	Low–med	"Not on the test" is the #1 complaint code (67 of 343)
6	**Bank-to-video pipeline**: Q&A practice videos per state and endorsement, in English and Spanish	Month 1	3	3	Medium	YouTube Q&A videos: Driving Academy GK 377k views (10 mo); 100% DMV 766k; DMV Advantage hazmat 61k (5 mo); Spanish: Rolando Cb 552k and 248k
7	**Endorsement completeness+**: all 6 endorsements, plus HazMat-renewal mode, L/Z air-brake restriction-removal path, CLP-to-skills checklist	Days 22–35	4–5	2	High (rivals have most)	Tanker/passenger demand above; HazMat retest at renewal in MO, LA, WY, PA, NY (state pages); TSA threat assessment every 5 yrs
8	**English-test bridge**: bilingual toggle (ES first), English term drill, roadside-English module, "your state tests in English" banner	Month 1–2	5–7	3	Medium	18% of truck drivers are foreign-born; Mexico 33% of them, 61% Hispanic (IIR/ACS 2018–22); FL and TX English-only; ELP roadside policy MC-SEE-2026-0002
9	**Pre-trip audio trainer**: walkaround order, call-out scripts, per-state modernized vs legacy	Month 2	4–5	3	Medium	Apex pre-trip video 8.4M views; Driving Academy 713k; Reddit "how to memorize pre-trip" threads; but only 17 of 6,112 reviews mention pre-trip (buyers are at the permit stage)
10	**Cab Mode**: voice-first, hands-free Q&A	Month 2	3	2	High	Solvely and CDL Pass (GP) already have audio
11	**Cited explanations**: every answer links to its manual section and page; precomputed, not a live chatbot	In MVP / wk 1	2–3	3 (with #2)	Medium	Hanh Le advertises an "AI Tutor" without citations; a live LLM has running cost against a one-time price
—	Pass guarantee	**Don't**	—	—	—	DMV Genie offers one with conditions. On iOS, Apple processes IAP refunds (DMV Genie terms), so we can't guarantee one. It's a claims and support burden that drifts toward service work
—	"Verified by retired examiner" review	**Option** (purchase,	1 + vendor	3	Medium	Price not researched; mark as an optional one-off spend after profit

#	Moat	When	CC-days [E]	Durability	Copyability	Key evidence
		needs Sean)				
—	Partnerships (schools, ELDT, WIOA, carriers)	**Defer: sales work**	—	4	Low	WIOA: 40% of 162k funded trainees in PY2024 were CDL (~65k [E]); FMCSA TPR 18,341 active providers. Every route needs outreach, so it breaks "no heavy sales"
—	Community / peer forum	Skip	—	1	—	Reddit r/Truckers and r/CDLTruckDrivers already fill it; moderation is a service-like cost

Sales work needed? Only partnerships (#—) and the examiner review (a purchase). Everything ranked 1–11 is build-only, with no outreach.

Moonbase check: nothing here is an agent-first or CX product, so none of it competes with Moonbase.

2. Coverage moat detail

State formats (sampled; full 51-row table is build task #1).

State	What's different	Source
Texas	Commercial Rules test (20 Q, 80%, taken first) → GK → Combination → Air Brake → endorsements. English-only since Jun 1, 2026 (Spanish dropped)	TX DPS apply page; 5-Star CLP packet; 37 TAC §16.27
New York	One correct answer from **three** choices; separate tests for each endorsement; HazMat state + TSA check	NY CDL-10 manual §1.4.1
Iowa	GK 50 Q, pass 40; three answer choices	Iowa DOT; driving-tests.org
California	GK 50 Q, 10 misses allowed	CA DMV evaluation report
Florida	English-only for all license exams incl. CDL since Feb 6, 2026	CDLLife, 2026
Oregon / Alabama / Maine / Pennsylvania	AAMVA modernized manual or tests: OR Oct 1, 2025 (old test to Jun 1, 2026); AL skills Jul 1, 2025; ME inspection Nov 1, 2026, full Sep 1, 2027; PA training Mar 2026	AAMVA; state notices

One discrepancy: cdlpracticetest.com says TX Commercial Rules is 25 Q, while DPS-linked material says 20. This is exactly the kind of error a sourced format table fixes.

Rival coverage sample (descriptions only, not verified in-app). - 8 of 11 iOS rivals claim 50 states: PATG, Hanh Le, Wei Zhang, Easy Pass, Khai Vuong, Solvely, BrainSoft, and DMV Genie ("1,500 state-specific questions"). - 0 of 11 mention TX Commercial Rules, three-choice formats, modernized manuals or English-only rules. - The only TX-specific product found is "TX CDL Driver Permit DMV test" (Google Play, Driver-Start), "based on the 2024 Texas manual", plus a web quiz at txcommercialrules.examzify.com. - Spanish: Hanh Le, Solvely, CDL Pass (GP). Audio: Solvely, CDL Pass. Pre-trip content: Easy Pass, DMV Genie, Solvely, BrainSoft. Guarantee: DMV Genie, GP "CDL Practice Written Test".

Adjacent licenses.

Adjacent	Verdict	Why
Car DMV permit	Month 3+, separate app, only if CDL clears the day-60 kill line	"dmv permit test" 33,100/mo but KD 59, owned by DMV Genie and Zutobi-class incumbents
Motorcycle permit	Possible month 3+ reuse of the engine	6,600/mo, KD 30
CDL / HazMat renewal	Build (cheap, in-bundle)	Renewal retest varies by state; "hazmat renewal test" only 40/mo, so it's a feature, not a channel
TWIC	Skip	It's a TSA background-check card with no knowledge test to study for

Adjacent	Verdict	Why
Forklift / OSHA	Skip	OSHA 1910.178 needs employer-run training and evaluation, which is compliance service territory

Ship speed [E]. - The state format table plus 51 state pages takes 6–8 CC-days, but human verification is the bottleneck: about 51 × 20 min ≈ 17 h of Sean or Lindsey checking against DMV sources. - Each extra endorsement bank takes about 1 CC-day to draft and 2–4 h to review. - Each extra language takes about 2 CC-days for the UI and glossary, plus a fluent reviewer (for Punjabi, Russian, Ukrainian, Somali or Amharic that's a paid reviewer, so do it only on data).

3. Data and feedback moat

- **How fast things change.** eCFR distinct amendment dates by year:

Part	2017–2025	2026 YTD	Last change
383	about 1–7 per year	8	Jul 22, 2026
172	about 1–10 per year	7	Sep 8, 2026
391	—	3	—
393	—	4	—

Add two PHMSA harmonization NPRMs in 2026 (HM-215R Feb 10; HM-215S Oct 27, comments due Dec 28, 2026), with a final rule likely in 2027 [E]. The modernized-manual rollout runs state by state through 2027. That works out to roughly a material change somewhere every 2–4 weeks [E]. - **Mechanics.** Each question stores source citations (CFR section or manual page). A nightly job diffs the eCFR versioner, the Federal Register API and DMV manual URLs, then flags every question citing a changed section. A fix goes out in the next build or an OTA update, and the log entry is public. - **Legal line.** Collect topic chips and "harder or easier than the app" ratings only. Never ask for, store or publish actual test questions. Many states treat test content as secure, and the user agreement should say so. Miss-rate data must be opt-in and aggregated, with no PII.

4. Distribution moats

- **SEO.**
- The 15-row Semrush batch adds about 4,650/mo across the state and endorsement terms (excluding the car and moto heads).
- With Task 1's terms, the CDL long tail is about 12k/mo at KD ≤ 28 [E].
- Spanish search demand is near zero: "cdl practice test spanish" shows 0 and "examen cdl en español" 50. Spanish demand lives on **YouTube**, not Google.
- Risk: Google's scaled-content policy. Every page needs unique, sourced state facts.
- **Video.** Q&A-style CDL test videos regularly reach 30k–770k views, from single-person channels. Some can be generated from the bank: question cards, voice-over and an answer reveal. No paid spend is needed, and no outreach. YouTube's repetitive-content rules mean each video needs real explanation, not just a slideshow.
- **Partnerships.** Sales work in every case, so defer them. The one no-outreach version is a public "for instructors" page with a free classroom quiz link.

5. Product moats (what rivals have)

Feature	Rivals with it	Our edge
Spanish	Hanh Le, Solvely, CDL Pass	English-test bridge (bilingual → English exam)
AI tutor	Hanh Le	Cited and precomputed; no running cost
Audio / Cab Mode	Solvely, CDL Pass	Hands-free full flows, not just read-aloud
Pre-trip	Easy Pass, Genie, Solvely, BrainSoft (text Q&A)	Audio walkaround, state modernized vs legacy
Readiness	Easy Pass	Calibrated with miss-rate data (month 2+)
Pass guarantee	DMV Genie	Don't copy

6. Brand and trust

- **Public accuracy log** from day 1. Every fix is dated and links the source and question ID.
- **Transparent changelog** in-app and on the web.
- **Reviews flywheel.** Prompt for a review after a passed mock and after the user reports passing the real test. Never prompt after a miss.
- **Timing [E].** At 80 sales/mo and about a 5% review rate, the app gets ~4 ratings a month, so ~50 ratings takes about a year. Speed can't shortcut this, which is why the log and SEO must carry trust early.
- **Examiner review.** Optional one-off purchase, after profit, with Sean's approval.

7. Where speed wins and where it doesn't

- **Week 1 post-MVP:** change-watch and fix log, cited explanations, TX Commercial Rules bank, format table for the top 10 states.
- **Month 1:** all 51 format rows and state pages, all endorsements plus renewal and L/Z, the video pipeline, test-day check-in.
- **Month 3:** miss-rate data, ES bridge, pre-trip audio, Cab Mode, motorcycle if the kill line has been cleared.
- **Speed doesn't help with:**
- App Review (1–3 days per build).
- The rating count (about a year to 50).
- Domain and page age (SEO typically takes 3–9 months [E]).
- Trust.
- Human verification of state facts.

8. Sharpened positioning

> **"The CDL test prep that matches your state's real test, fixed in public when the rules change. Every endorsement, $34.98 once. No ads, no auto-renew."**

- App Store subtitle: "Your state's exact CDL test."
- Proof points: format card, fix log, cited answers.

9. 90-day roadmap after the 21-day MVP

Days	Ship	Gate
22–35	Change-watch + /fixes log; cited explanations on all Qs; TX Commercial Rules bank; format table + pages for top 10 states (TX, CA, FL, GA, IL, PA, OH, NY, NC, IN [JJ]); tanker/passenger/school bus/doubles pages	Fix SLA ≤ 7 days
36–60	All 51 rows + pages; HazMat renewal + L/Z modes; video pipeline (2 videos/wk EN, 1 ES); test-day topic check-in	**Day 60 kill line: under 25 sales/mo means stop**
61–90	Miss-rate dashboard + "hardest questions" pages; ES English-test bridge; pre-trip audio v1	≥ 50 sales/mo to continue the heavy build
91–111	Cab Mode; readiness recalibration; motorcycle reuse (only if ≥ 80 sales/mo)	Target 80/mo

10. Odds impact (model in `model/cdl_moat_run.py`, log `cdl_moat_log.txt`)

The patched engine changes only app #2 (CDL); all other apps stay at 0.12. Scenario B, C replaces CAL, N=200k, seed 1027. The 0.12 row reproduces the Oct 28 numbers exactly.

CDL p1k	Base m12	Base m18	Pess m12	Pess m18
0.12 (default)	67.1%	81.3%	61.4%	77.5%
0.15	67.3%	81.4%	61.6%	77.6%
0.18 [J]	**67.5%**	**81.5%**	**61.9%**	**77.7%**

CDL p1k	Base m12	Base m18	Pess m12	Pess m18
0.22	67.9%	81.6%	62.2%	77.9%
0.30	68.5%	81.9%	63.0%	78.2%

Single CDL app, monthly net at maturity:

p1k	Median	p75	p90	P(≥$1k)	P(≥$3k)
0.12	$74	$324	$1,240	11.8%	4.6%
0.18	$131	$575	$2,188	17.8%	7.8%
0.22	$178	$785	$2,988	21.6%	10.0%

Why 0.18 and not higher [J]. - For: proven paid demand (top 4 apps ~$205k/mo [E]); 33% of complaints are accuracy, which our moats target directly; a low-KD long tail; a record that compounds. - Against: 8 of 11 rivals already claim 50 states; Hanh Le is polished and has AI and Spanish; English-only testing and the non-domiciled rule shrink the immigrant segment; ratings take about a year.

Bottom-up check [E]. At an average ticket of about $22 (only an estimate) and Apple's 15% small-business rate: - The 80-sales target is about $1.5k/mo net. - The 25-sales kill line is about $470/mo. - $1k/mo net needs about 54 sales/mo.

11. Spend

- Semrush: 15 rows (150 units).
- Apify: $0 this task (no new runs).
- Treg: $0.
- Everything else came from free sources: eCFR API, YouTube search page, web search, saved store data.

12. Sources (main)

- TX DPS CDL page and English-only notice; 37 TAC §16.27; 5-Star CLP packet PDF.
- NY CDL-10 manual §1.4.1; Iowa DOT CDL testing; CA DMV CDL knowledge test evaluation.
- FMCSA States page (49 CFR 383 Subpart G/H); FMCSA ELP roadside policy MC-SEE-2026-0002; FMCSA non-domiciled CDL final rule and FAQs; TTNews (28k revoked).
- CDLLife (Florida English-only, 2026).
- eCFR versioner API (raw: `moat/ecfr_v_*.json`).
- Federal Register HM-215R (91 FR 5996, Feb 10, 2026) and HM-215S (Oct 27, 2026).
- AAMVA CDL testing page; Oregon DMV bulletin; ALEA; Maine SOS.
- TSA HME pages; MO DOR HME FAQ; LA OMV renewal.
- IIR/GMU Immigrant Truck Drivers 2022 (ACS 2018–22 via IPUMS); NATSO.
- retrainingreport.com (WIOA PY2024); FMCSA TPR dashboard.
- DMV Genie premium and terms pages.
- Semrush (raw: `moat/semrush_15.csv`); YouTube search parse (raw: `moat/youtube_search.json`).
- App Store and Google Play descriptions (`pricing/*.json`); UX report coded complaints (n=343).